

# Nikolas Dinh

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<https://nikolasdinh.github.io/>

## EDUCATION

### California Polytechnic University, Pomona

*Mechanical Engineering Student 3.5 GPA*

Expected Graduation Class of 2027

## SKILLS

**CAD:** Solidworks (**CSWP**), AutoCAD, Fusion 360, OnShape

**Manufacturing:** 3D Printing, Laser Cutting, Wood Working, GD&T, Prototyping, Welding (MIG, TIG, Stick, Flux Core)

**Electronics:** Arduino, Raspberry Pi, Soldering, Wiring Schematics

**Programming:** MATLAB, Ansys Fluid Flow Fluent, Python, C++

## PROJECTS

### FSAE Beam Wing

Oct 2025 - Dec 2025

- Achieved net ~5 lb increase in downforce between beam wing and undertray improvements and rearward center-of-pressure shift; identified wake interference as a limiting factor
- Researched NACA airfoil geometries to design a rear beam wing for CPP FSAE maintaining a 6 week R&D timeline
- Conducted 21+ CFD simulations with **SolidWorks** and in **ANSYS Fluent** to evaluate downforce, wake behavior, and aerodynamic efficiency with drag considering other aerodynamic tools
- Determined addition of a beam wing would be incoherent with the overall goals and cost of the current aero package

### CNC Router

Apr 2026-May 2026

- Designed and built an autonomous CNC drill prototype using **SolidWorks**, **Arduino**, NEMA 17 stepper motors, belts, lead screws, and aluminum extrusions under a \$200 budget.
- Developed X/Y/Z motion systems to drill required hole locations, depths, and diameters into a foam block.
- Created CAD assemblies and engineering drawings for the carriage, block holder, Z-axis frame, and spindle components.
- Programmed Arduino motion commands to complete autonomous drilling and swarf removal, achieving all required holes in approximately 1 minute 40 seconds.

### HVAC Cost Modeling

Jan 2025 - Feb 2025

- Collected and filtered 1000+ data points measures of lux, temperature, and insulation with Vernier probing
- Predicted future thermal behavior in California using collected data and NOAA average temperatures in **MATLAB**
- Modeled cost of running air conditioning to maintain comfortable temperatures with a \$32.68 per KW for 365 days

### Knife Design and Fabrication

March 2025 - June 2025

- Engineered a fixed-blade knife using A2 tool steel, heat treatment to achieve a post-temper hardness of 51.5 HRC and maximum bending load of 13.65 KN using **Universal Instron**
- Modeled with high saber grind and custom PLA Wa-style handle, friction fit sheath using **SolidWorks** and **3D printing**
- Conducted full fabrication cycle including beveling, tempering, and sharpening using hand tools and wet stone from McMaster-Carr stock material, achieving the fastest slicing time (0.79 s) in the final evaluation.

### Winch Sled

Jan 2026 - Mar 2026

- Led team design in winch sled pulling 4 circular 500g masses a height of 5ft, distance of 30in in 1.43sec staying \$23 under budget and 6V DC battery limitation
- Researched battery efficiency in combination with servo performance for a 2:1 mechanical advantage pulley system
- Designed sled to pull with a flywheel with weights loaded in a monorail carriage up 2x4 pine

### Project Car Cost Program

Oct 2025 - Dec 2025

- Developed **Python** script program in **Visual Studio Code** to organize 38 data points and \$16,000 worth of cost
- Clean UI interface with tkinter and DB browser for data management for ease of use limiting complexity
- Organizes tabs for 14 customers at automechanic shop improving transparency and confidence

## WORK EXPERIENCE

### VSE Corporation — Manufacturing Engineering Intern

Summer 2026

- Accepted upcoming internship supporting manufacturing process improvement, production operations, and continuous improvement initiatives.
- Expected to work with engineering drawings, tooling, workflows, SOPs, work instructions, and process documentation.
- Selected to apply SolidWorks/CAD experience, blueprint interpretation, and engineering design knowledge in a manufacturing environment.